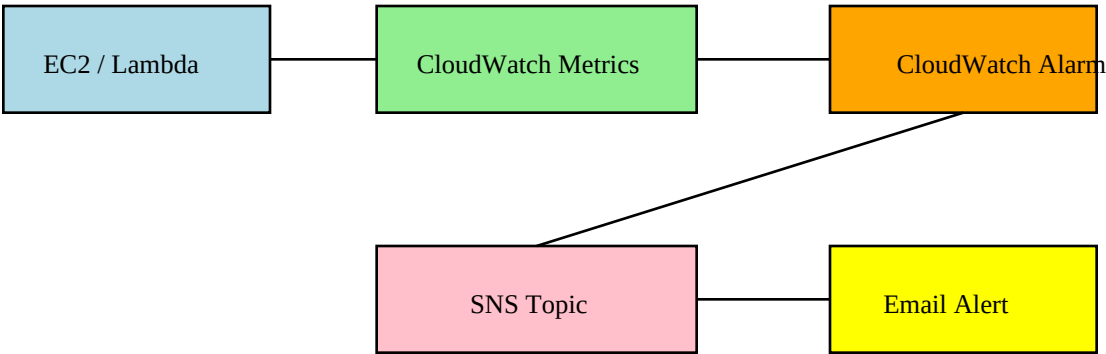


AWS CloudOps Engineering Associate – Day 8

Amazon CloudWatch Monitoring & Logging

Amazon CloudWatch is a monitoring and observability service in AWS that helps cloud engineers track infrastructure health, monitor applications, collect logs, create alarms, and automate responses. It plays a major role in CloudOps and DevOps environments.

CloudWatch Monitoring Workflow Diagram



Workflow Explanation:

1. AWS resources such as EC2, Lambda, or RDS generate metrics.
2. Amazon CloudWatch continuously monitors these metrics.
3. When a threshold is crossed (example: CPU > 80%), a CloudWatch Alarm is triggered.
4. The alarm sends a notification to an SNS Topic.
5. SNS delivers alerts through Email, SMS, or automation workflows.

Core CloudWatch Features

Feature	Purpose
Metrics	Track CPU, memory, network usage
Logs	Store application and system logs
Alarms	Trigger notifications automatically
Dashboards	Visualize infrastructure health
Log Insights	Analyze logs using queries

Hands-On Practice:

- Launch an EC2 instance.
- Open CloudWatch Console.
- Monitor CPUUtilization metric.
- Create a CloudWatch Alarm for CPU > 70%.
- Configure SNS Email Notifications.
- Perform stress testing to trigger the alarm.

Best Practices:

- Enable detailed monitoring for production workloads.
- Create alarms for critical infrastructure metrics.
- Configure log retention policies.
- Use dashboards for centralized visibility.
- Integrate CloudWatch with Lambda and EventBridge for automation.

Conclusion:

Amazon CloudWatch is one of the most important AWS services for CloudOps engineers. It provides monitoring, logging, alerting, and observability capabilities that help maintain high availability and performance in cloud environments.